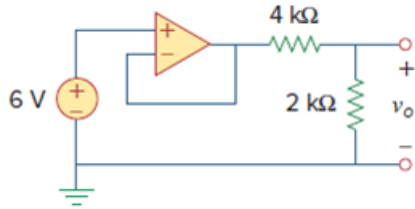


Problem

The power absorbed by the $4\text{-k}\Omega$ resistor in Fig. 5.42 is:

- (a) 9 mW
- (b) 4 mW
- (c) 2 mW
- (d) 1 mW

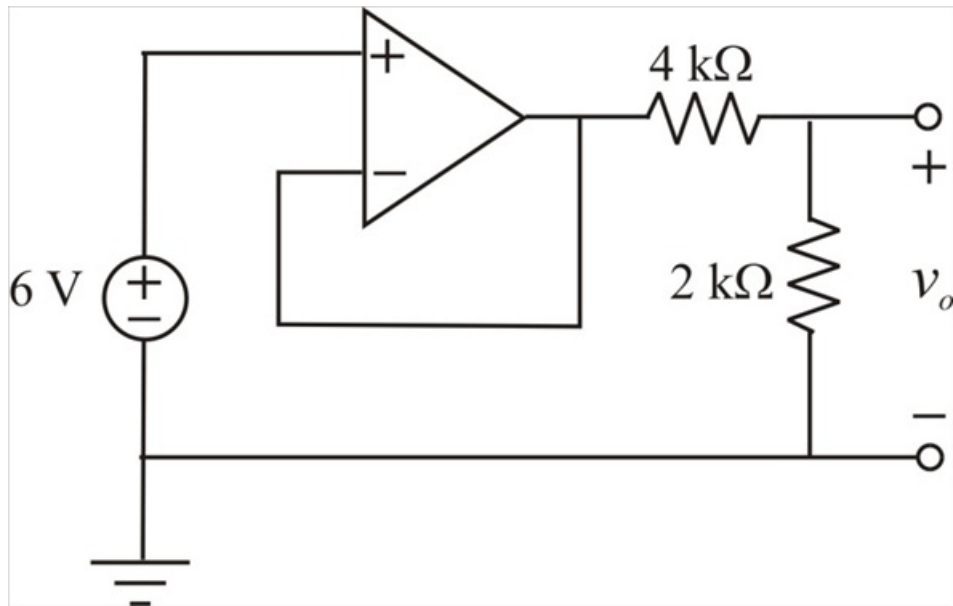
Figure. 5.42:



Step-by-step solution

Step 1 of 7

Refer to Figure 5.42 in the text book.



Step 2 of 7

Redraw the circuit as shown in Figure 1.

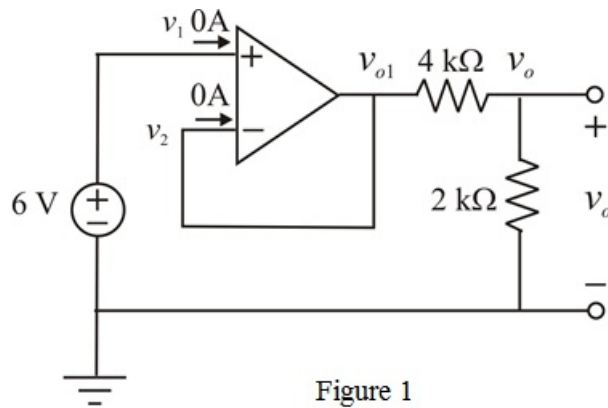


Figure 1

Step 3 of 7

From the circuit shown in Figure 1,

$$v_1 = 6 \text{ V and}$$

$$v_2 = v_{o1}.$$

In an ideal op-amp, the inverting and non-inverting terminal currents are zero. And the inverting and non-inverting node voltages are equal.

i.e.,

$$v_1 = v_2$$

$$v_1 = v_2 = 6 \text{ V} \quad (\text{Since, } v_1 = 6 \text{ V})$$

$$v_{o1} = 6 \text{ V} \quad (\text{Since, } v_2 = v_{o1})$$

Hence, the voltage $v_{o1} = 6 \text{ V}$.

Step 4 of 7

Apply voltage division rule to the circuit shown in Figure 1.

$$v_o = \left(\frac{2 \times 10^3}{2 \times 10^3 + 4 \times 10^3} \right) v_{o1}$$

$$v_o = \left(\frac{2}{6} \right) v_{o1}$$

Substitute 6 V for v_{o1} in the equation.

$$v_o = \left(\frac{2}{6} \right) (6)$$

$$v_o = 2 \text{ V}$$

Therefore, $v_o = 2 \text{ V}$.

Step 5 of 7

Calculate the power absorbed by the $4\text{ k}\Omega$ resistor.

$$\begin{aligned}P_{4\text{ k}\Omega} &= \frac{(v_{oi} - v_o)^2}{4 \times 10^3} \\&= \frac{(6 - 2)^2}{4 \times 10^3} \\&= \frac{(4)^2}{4 \times 10^3} \\&= 4 \times 10^{-3} \text{ W} \\P_{4\text{ k}\Omega} &= 4 \text{ mW}\end{aligned}$$

Step 6 of 7

The power absorbed by the $4\text{ k}\Omega$ resistor is not 9 mW , 2 mW and 1 mW . Hence, the option (a), (c) and (d) are incorrect.

Step 7 of 7

The power absorbed by the $4\text{ k}\Omega$ resistor, $P_{4\text{ k}\Omega}$ is 4 mW .

Hence, Option **(b)** is correct.